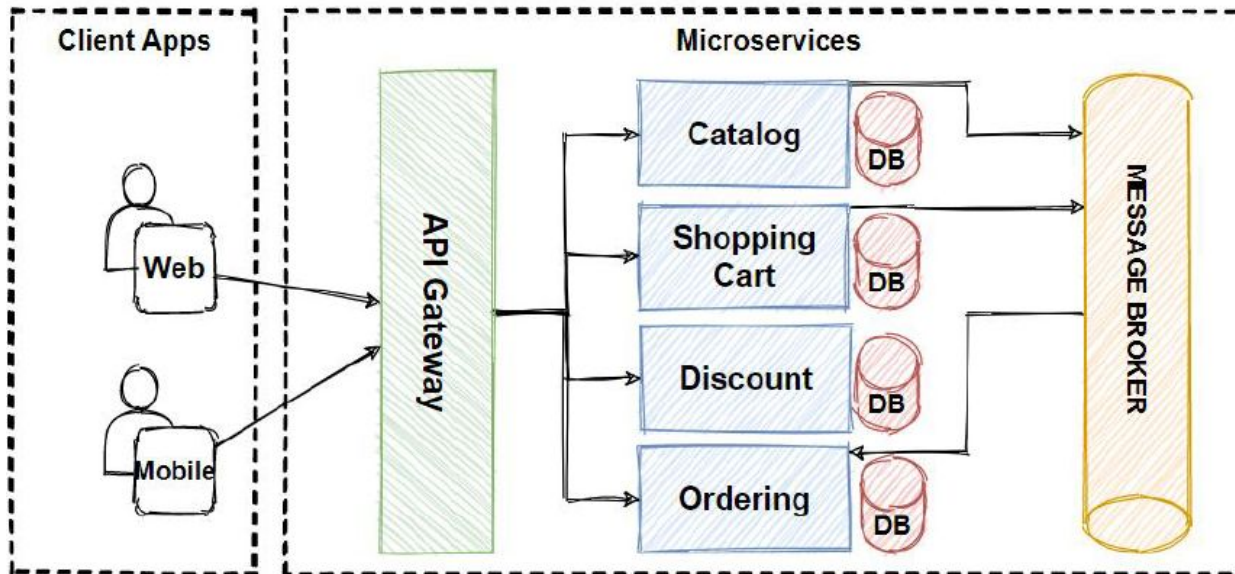


How Reliable Are LLM Evaluations of Microservice Architecture Diagrams?

João Luiz Cerqueira, Mateus Dutra, Eduardo Figueiredo,
Gustavo Vale

Context & Research Problem

- Software Architecture diagrams express service boundaries, API gateways, message brokers, and database ownership in microservices.



Context & Research Problem

- Rapidly evolving systems cause diagrams to become obsolete, incomplete, ambiguous, or overly informal.
- Multimodal LLMs emerge as potential automated reviewers, but their empirical reliability on visual diagrams was unknown.

Background

Baseline Study: Oliveira & Mendonça (ECSA 2025)

"LLM-Based Quality Assessment of Software Architecture Diagrams: A Preliminary Study with Four Open-Source Projects"

Evaluated the feasibility of a single LLM (ChatGPT-4o) on 4 open-source architecture diagrams using 5 criteria (Accuracy, Clarity, Completeness, Consistency, Detail)

Goal and Research Questions

- **Goal:** Evaluate the reliability of LLM-based microservice architecture diagram assessments by analyzing score alignment against human reference scores and expert preferences for generated reasoning.

Goal and Research Questions

- **RQ1:** To what extent do LLM-generated scores align with human reference scores when assessing microservice architecture diagrams?
- **RQ2:** Which quality criteria show stronger or weaker agreement between LLM-generated scores and human reference scores?
- **RQ3:** Do software engineers prefer the reasoning generated from image-based or text-based diagram representations?

Goal and Research Questions

1. Data Collection

6 real-world open-source microservice systems. PNG, SVG/XML, and text descriptions.

3. Human Audit

16 Software Engineers.
12 balanced human reference sets (ground truth) + 253 pairwise blind justification comparisons.

2. LLM Evaluation

GPT-5 and Claude Sonnet 4.6.
2 representations (PNG vs SVG/XML). 5 criteria, 15 guidelines. 120 API calls

2. Data Analysis

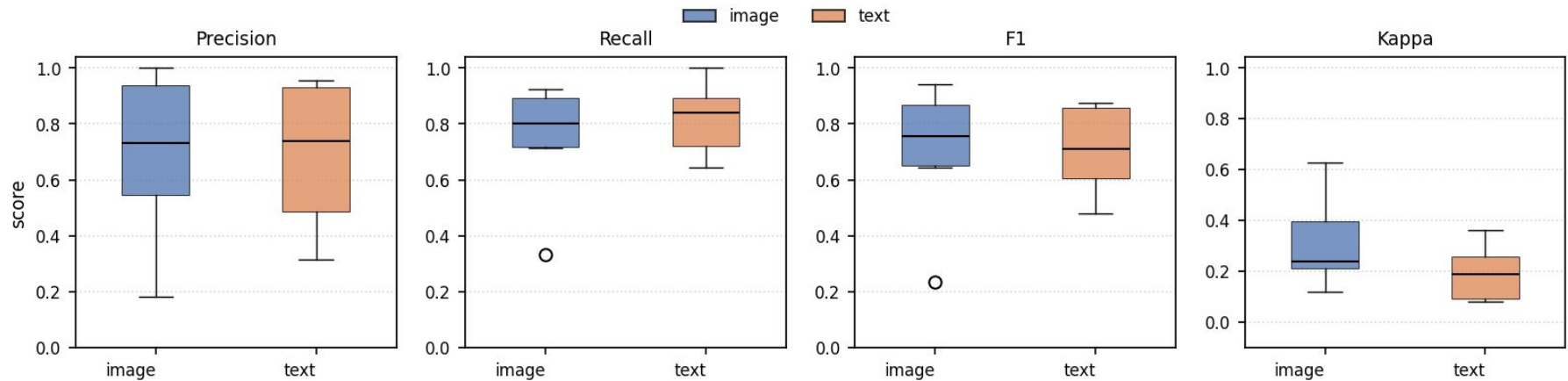
F1, Precision, Recall, Quadratic Weighted Kappa

5 criteria + 15 Guidelines

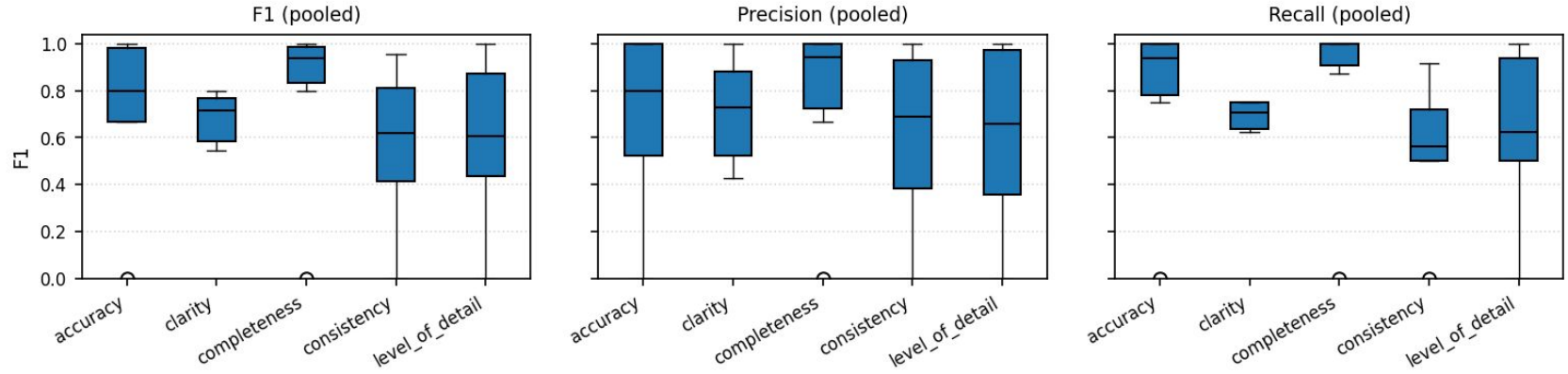
Quality Criteria	Definition	Guideline-Level Checks
<i>Accuracy</i>	Does the diagram correctly reflect the architecture described in text?	1. Component matching 2. Interaction matching
<i>Clarity</i>	Can a technical reader easily understand the diagram and relationships?	3. Element identification 4. Reading flow 5. Label readability 6. Specific symbology
<i>Completeness</i>	Can a technical reader easily understand the diagram and relationships?	7. Coverage of main components 8. External actors coverage 9. Main interactions coverage
<i>Consistency</i>	Does it apply notation and technology symbols uniformly?	10. Notation definition & uniformity 11. Technology symbols 12. Naming consistency
<i>Level of Detail</i>	Is the level of detail appropriate for developers and architects?	13. Element detail 14. Interaction detail 15. Overall appropriateness

Scoring scale: 4 = Excellent, 3 = Good, 2 = Fair, 1 = Poor

Results: RQ1



Results: RQ2



Strong Suit: Information Presence

LLMs excel in **Completeness** (Recall 0.975) and **Accuracy** (Recall 0.938).

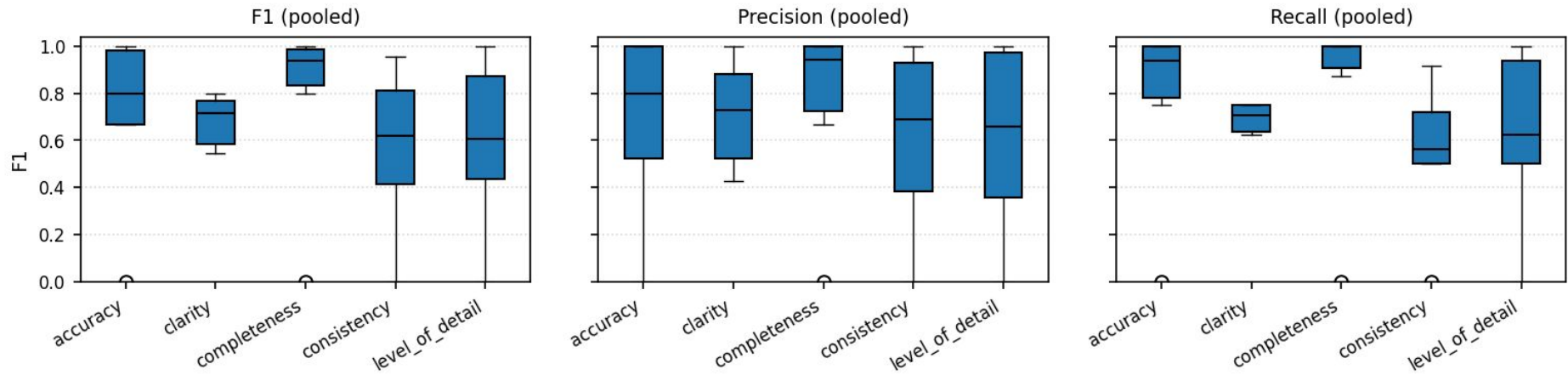
Outstanding performance when checking checklist-style component presence/absence.

Weak Point: Subjective Interpretation

LLMs underperform in **Clarity** (lowest F1: 0.697, Recall: 0.679) and **Level of Detail**.

Difficulty evaluating visual layout readability and audience expectations.

Results: RQ3



Key Findings

- LLMs should not act as authoritative evaluators. They function best as first-pass triage filters to flag incomplete or suspect diagrams for human inspection.
- Decomposing evaluation into 15 concrete guidelines converts generic criticisms ("diagram is poor") into actionable, auditable feedback items.
- The optimal workflow combines automated LLM triage with human expert validation, especially for subjective criteria like clarity and level of detail.

Thank you!
Questions?